Lesson Planning

Complete Lesson Plan Book

Class: Class 6

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Integers: Positive and Negative

Learning Objectives

• Students will be able to define integers with examples.

• Students will be able to place integers on a number line.

• Students will be able to add and subtract simple integers.

Key Concepts

• Integers: ..., -3, -2, -1, 0, 1, 2, 3, ...

• Positive numbers are greater than zero; negative are smaller.

• Number line: negatives left, positives right.

• Adding integers with the same and different signs.

Teaching Methodology (Step by Step)

1. Begin with a 5-minute warm-up: ask 2-3 oral questions revising the previous lesson and linking it to Integers: Positive and Negative.

2. Present the key ideas of Integers: Positive and Negative on the board with one clear example or a labelled diagram.

3. Explain each point slowly in simple words, asking a short check-question after every point.

4. Do guided practice: solve 3-4 examples of Integers: Positive and Negative together, inviting students to complete steps on the board.

5. Close with a 5-minute oral recap in which students state one new thing they learned about Integers: Positive and Negative.

Classroom Activities

• Number line walk: students step left and right on a floor number line.

• Integer cards: add pairs of integer cards in teams.

Assessment Questions

1. Write 5 integers: 3 positive and 2 negative.

2. Add: (-4) + 7.

3. Subtract: 3 - 8.

Homework

• Draw a number line and mark -5 to +5; solve 6 integer sums.

Time Allocation

Two periods (80 minutes): 10 min warm-up and revision, 25 min concept presentation, 25 min guided and independent practice, 15 min activity and assessment, 5 min closure and homework.

Unit 2: Fractions: Addition and Subtraction

Learning Objectives

• Students will be able to add like fractions.

• Students will be able to add unlike fractions using LCM.

• Students will be able to subtract fractions and simplify answers.

Key Concepts

• Like fractions: same denominator.

• LCM of denominators for unlike fractions.

• Adding and subtracting mixed numbers.

• Simplifying the final answer.

Teaching Methodology (Step by Step)

1. Start with rapid-fire mental revision: 5 quick questions from the last lesson connected to Fractions: Addition and Subtraction.

2. Demonstrate Fractions: Addition and Subtraction step by step on the board while students watch, listen and copy.

3. Move to guided practice: students attempt 2-3 similar items with the teacher's support.

4. Give independent practice: a short written task on Fractions: Addition and Subtraction completed individually in notebooks.

5. Check a few notebooks on the spot, correct common mistakes on the board and summarise Fractions: Addition and Subtraction.

Classroom Activities

• Fraction strips: students add strips to see sums visually.

• LCM race: teams find LCMs of denominator pairs.

Assessment Questions

1. Add: 3/8 + 2/8.

2. Add: 1/2 + 1/3.

3. Subtract: 5/6 - 1/4 and simplify.

Homework

• Solve 6 mixed fraction addition and subtraction sums.

Time Allocation

Two periods (80 minutes): 10 min revision, 20 min explanation with examples, 30 min practice and class activity, 15 min oral and written check, 5 min homework briefing.

Unit 3: Decimals: Multiplication and Division

Learning Objectives

• Students will be able to multiply decimals by whole numbers.

• Students will be able to divide decimals by whole numbers.

• Students will be able to place the decimal point correctly in answers.

Key Concepts

• Multiplying: ignore point, then count decimal places.

• Dividing: divide as whole numbers, place point above.

• 0.5 x 4 = 2.0; 7.5 / 3 = 2.5.

• Word problems with money decimals.

Teaching Methodology (Step by Step)

1. Open with a hook: a short story, riddle or real-life example that introduces Decimals: Multiplication and Division.

2. Lead a class discussion on Decimals: Multiplication and Division with 4-5 guiding questions; record key answers on the board.

3. Organise the main activity in pairs or small groups according to the plan for Decimals: Multiplication and Division.

4. Let groups share their work; add corrections and the important points of Decimals: Multiplication and Division.

5. Finish with a quick oral or written check on Decimals: Multiplication and Division and announce the homework.

Classroom Activities

• Money maths: students solve shop problems with decimal prices.

• Point placers: students fix deliberately misplaced decimal points.

Assessment Questions

1. Multiply: 2.4 x 3.

2. Divide: 8.4 / 4.

3. A pen costs Rs 12.50. What is the cost of 6 pens?

Homework

• Solve 5 multiplication and 5 division decimal sums.

Time Allocation

Two periods (80 minutes): 10 min warm-up and revision, 25 min concept presentation, 25 min guided and independent practice, 15 min activity and assessment, 5 min closure and homework.

Unit 4: Ratio and Proportion

Learning Objectives

• Students will be able to define ratio with examples.

• Students will be able to simplify ratios to lowest terms.

• Students will be able to solve simple proportion problems.

Key Concepts

• Ratio compares two quantities: 2:3.

• Simplifying: 6:9 = 2:3.

• Proportion: two equal ratios.

• Unitary method for proportion problems.

Teaching Methodology (Step by Step)

1. Begin with 5 minutes of silent or loud reading connected to Ratio and Proportion.

2. Explain the new words and key ideas of Ratio and Proportion; write difficult words with meanings on the board.

3. Read the passage or examples of Ratio and Proportion together, with students taking turns.

4. Students complete a short written exercise on Ratio and Proportion in their notebooks.

5. Review the answers together, praise good work and give a brief summary of Ratio and Proportion.

Classroom Activities

• Mixture activity: students mix juice and water in given ratios.

• Ratio hunt: find ratios in classroom objects, e.g. boys to girls.

Assessment Questions

1. Simplify the ratio 12:18.

2. If 3 pencils cost Rs 30, what is the cost of 7 pencils?

3. Write the ratio of 20 boys to 15 girls in simplest form.

Homework

• Solve 5 ratio and 3 proportion word problems.

Time Allocation

Two periods (80 minutes): 10 min revision, 20 min explanation with examples, 30 min practice and class activity, 15 min oral and written check, 5 min homework briefing.

Unit 5: Percentage

Learning Objectives

• Students will be able to define percent as per hundred.

• Students will be able to convert fractions to percentages.

• Students will be able to find the percentage of a quantity.

Key Concepts

• Percent means out of 100; symbol %.

• Fraction to percent: multiply by 100.

• Finding x% of a number.

• Daily uses: marks, discounts, battery.

Teaching Methodology (Step by Step)

1. Begin with a 5-minute warm-up: ask 2-3 oral questions revising the previous lesson and linking it to Percentage.

2. Present the key ideas of Percentage on the board with one clear example or a labelled diagram.

3. Explain each point slowly in simple words, asking a short check-question after every point.

4. Do guided practice: solve 3-4 examples of Percentage together, inviting students to complete steps on the board.

5. Close with a 5-minute oral recap in which students state one new thing they learned about Percentage.

Classroom Activities

• Discount shop: students calculate sale prices with 10% and 20% off.

• Marks to percent: convert test marks into percentages.

Assessment Questions

1. Convert 3/4 into a percentage.

2. Find 25% of 200.

3. A shirt costs Rs 800 with 10% discount. What is the sale price?

Homework

• Convert 5 fractions to percent; find percentages of 6 quantities.

Time Allocation

Two periods (80 minutes): 10 min warm-up and revision, 25 min concept presentation, 25 min guided and independent practice, 15 min activity and assessment, 5 min closure and homework.

Unit 6: Algebra: An Introduction

Learning Objectives

• Students will be able to explain what a variable is.

• Students will be able to write algebraic expressions for word statements.

• Students will be able to evaluate expressions for given values.

Key Concepts

• Variable: a letter standing for a number.

• Expression: e.g. x + 5, 3y - 2.

• Translating words: 5 more than n = n + 5.

• Evaluating: if x = 4, then 2x + 1 = 9.

Teaching Methodology (Step by Step)

1. Start with rapid-fire mental revision: 5 quick questions from the last lesson connected to Algebra: An Introduction.

2. Demonstrate Algebra: An Introduction step by step on the board while students watch, listen and copy.

3. Move to guided practice: students attempt 2-3 similar items with the teacher's support.

4. Give independent practice: a short written task on Algebra: An Introduction completed individually in notebooks.

5. Check a few notebooks on the spot, correct common mistakes on the board and summarise Algebra: An Introduction.

Classroom Activities

• Expression match: match word statements with algebraic expressions.

• Value machine: students evaluate expressions for given variable values.

Assessment Questions

1. Write the expression: 7 less than p.

2. If a = 5, find the value of 3a + 2.

3. Write in words: 4m - 6.

Homework

• Write expressions for 5 word statements; evaluate 5 expressions.

Time Allocation

Two periods (80 minutes): 10 min revision, 20 min explanation with examples, 30 min practice and class activity, 15 min oral and written check, 5 min homework briefing.

Unit 7: Triangles and Quadrilaterals

Learning Objectives

• Students will be able to classify triangles by sides and angles.

• Students will be able to name types of quadrilaterals.

• Students will be able to state the angle sum of a triangle.

Key Concepts

• Triangles: equilateral, isosceles, scalene; acute, right, obtuse.

• Quadrilaterals: square, rectangle, parallelogram, rhombus, trapezium.

• Angles of a triangle add to 180 degrees.

• Properties of each quadrilateral.

Teaching Methodology (Step by Step)

1. Open with a hook: a short story, riddle or real-life example that introduces Triangles and Quadrilaterals.

2. Lead a class discussion on Triangles and Quadrilaterals with 4-5 guiding questions; record key answers on the board.

3. Organise the main activity in pairs or small groups according to the plan for Triangles and Quadrilaterals.

4. Let groups share their work; add corrections and the important points of Triangles and Quadrilaterals.

5. Finish with a quick oral or written check on Triangles and Quadrilaterals and announce the homework.

Classroom Activities

• Triangle sort: classify cut-out triangles by sides and angles.

• Quadrilateral walk: students form shapes with arms and bodies.

Assessment Questions

1. Classify a triangle with sides 5 cm, 5 cm, 8 cm.

2. Name two differences between a square and a rhombus.

3. Two angles of a triangle are 60 and 70 degrees. Find the third.

Homework

• Draw and label 3 triangles and 4 quadrilaterals with properties.

Time Allocation

Two periods (80 minutes): 10 min warm-up and revision, 25 min concept presentation, 25 min guided and independent practice, 15 min activity and assessment, 5 min closure and homework.

Unit 8: Perimeter, Area and Volume

Learning Objectives

• Students will be able to find the perimeter of squares and rectangles.

• Students will be able to find the area of squares and rectangles.

• Students will be able to find the volume of a cuboid.

Key Concepts

• Perimeter: sum of all sides.

• Area of rectangle = length x width; square = side x side.

• Area of triangle = half x base x height.

• Volume of cuboid = length x width x height.

Teaching Methodology (Step by Step)

1. Begin with 5 minutes of silent or loud reading connected to Perimeter, Area and Volume.

2. Explain the new words and key ideas of Perimeter, Area and Volume; write difficult words with meanings on the board.

3. Read the passage or examples of Perimeter, Area and Volume together, with students taking turns.

4. Students complete a short written exercise on Perimeter, Area and Volume in their notebooks.

5. Review the answers together, praise good work and give a brief summary of Perimeter, Area and Volume.

Classroom Activities

• Measure it: students measure desk top and find its perimeter and area.

• Box volume: fill small boxes with cubes to find volume.

Assessment Questions

1. Find the perimeter of a rectangle 8 cm by 5 cm.

2. Find the area of a square of side 9 cm.

3. Find the volume of a cuboid 4 cm x 3 cm x 2 cm.

Homework

• Solve 3 perimeter, 3 area and 2 volume problems.

Time Allocation

Two periods (80 minutes): 10 min revision, 20 min explanation with examples, 30 min practice and class activity, 15 min oral and written check, 5 min homework briefing.